^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18

1972Fr11: 3.50- and 4.00-MeV deuteron beams were produced from the 4-MV Van de Graaff accelerator at the Nuclear Research Center of Strasbourg, France. Targets were antimony sulphide enriched to 85.6%, 550 μg/cm^2 on a 0.03 mm molybdenum layer and 150 μg/cm^2 on a 0.05 mm silver layer. Protons were detected using a surface-barrier annular silicon detector at 180°. γ rays were detected using a 40 cm^3 Ge(Li) detector at 0°, 55° and 90° with FWHM=3.2 keV at Eγ=1.332 MeV in the first experiment and a NaI crystal at 0°, 30°, 45°, 60° and 90° in the second experiment. Measured E_p, Eγ, Iγ, and p$\gamma(\theta)$-coin. Deduced 12 levels, J, γ-ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 7 levels using the Doppler-Shift Attenuation Method (DSAM).

1975VaYG,1972Va07: A 4.72-MeV deuteron beam of 175 nA was provided from the Groningen 5 MV Van de Graaff generator. Targets were 80-μg/cm^2 Zn^{34}S enriched to 89% evaporated onto 10-μg/cm^2 Formvar plus 10-μg/cm^2 carbon. Protons were detected using a 2-mm annular silicon detector. γ rays were detected using a 7.6-cm×7.6-cm NaI(Tl) detector. Measured E_p, Eγ, Iγ, and p$\gamma(\theta)$-coin. Deduced levels, J, γ-ray branching ratios, and mixing ratios. Also see 1971VaZG.

1970Bu18: 2-3.25 MeV deuteron beams were produced from the University of Arizona 5.5 MV Van de Graaff accelerator. Targets were 480-μg/cm^2 *120* Ag_2S made by heating 67% enriched ^{34}S powder and silver foil. Protons were detected using a 1-mm annular silicon surface barrier detector and γ rays were detected using a 7.62 cm×7.62 cm NaI(Tl) detector in the angular correlation experiment and a Ge(Li) detector in the lifetime experiment. Measured E_p, Eγ, Iγ, and p$\gamma(\theta)$-coin. Deduced 3 levels, J, γ-ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 3 levels using the Doppler-shift attenuation method (DSAM).

2024Co04: A 16-MeV deuteron beam from the John D. Fox Superconducting Linear Accelerator, FSU. Particles were momentum-analyzed by the Super-Enge Split-Pole Spectrograph and identified using a position-sensitive ionization chamber with two anode wires in the isobutane gas (ΔE) and a large plastic scintillator (E) at the focal plane. γ rays were detected using a $CeBr_3$ Array of five $CeBr_3$ detectors. Measured pγ delayed coincidence time spectra and deduced $T_{1/2}$ for the 1991 level of ^{35}S.

1971Pr11: A 4.48-MeV deuteron beam was produced from the insulated-core transformer tandem accelerator at the Aerospace Research Laboratories (ARL). The target was Ag_2S enriched to 85.61% in ^{34}S. Protons were detected using a 500 μm-thick, 50-mm^2 silicon detector and γ rays were detected using a 1-in.-thick, 1.5-in. diameter NE111 scintillator. Measured pγ-delayed coincidence time spectrum and deduced $T_{1/2}$ for the 1.99-MeV level.

1972Go31: A 3.3-MeV deuteron beam was produced from the BNL 3.5-MeV van de Graaff facility. The target was Sb_2S_3 enriched to 86% in ^{34}S. γ rays were detected using a Ge(Li) detector. Measured Eγ=1572.24 *17* deexciting the first excited state.

^{35}S Levels

E(level)[†]	J$^\pi$[‡]	$T_{1/2}$	Comments
0	3/2$^+$		
1572.33 *20*	1/2$^+$	>1.3 ps	E(level): weighted average of 1574.2 *10* (1970Bu18), 1572.3 *3* (1972Fr11), and 1572.28 *17* (1972Go31). J$^\pi$: spin=1/2 from isotropic $\gamma(\theta)$ in 1972Fr11 and 1972Va07. $T_{1/2}$: from τ>1.8 ps in 1970Bu18 by DSAM. Other: >1.1 ps (1972Fr11, τ>1.6 ps).
1992.9 *17*	7/2$^-$	1.03 ns *5*	E(level): unweighted average of 1994.6 *10* (1970Bu18) and 1991.2 *3* (1972Fr11). J$^\pi$: spin=5/2, 7/2 from $\gamma(\theta)$ in 1972Fr11, 1972Va07, and 1975VaYG. $T_{1/2}$: from τ=1.48 ns *7* from weighted average of 1.7 ns *3* (2024Co04,) and 1.47 ns *7* (1971Pr11), both from pγ(t). Other: τ>4.5 ps using DSAM (1970Bu18).
2348.8 *13*	3/2$^-$	0.71 ps *15*	E(level): unweighted average of 2350.0 *10* (1970Bu18) and 2347.5 *2* (1972Fr11). J$^\pi$: spin from $\gamma(\theta)$ in 1972Va07, 1975VaYG, and 1972Fr11. $T_{1/2}$: from τ=1.03 ns *21* from weighted average of 0.8 ps *2* (1970Bu18) and 1.22 ps *18* (1972Fr11), both by DSAM.
2718.0 *10*	5/2$^+$	69 fs *24*	J$^\pi$: spin=3/2, 5/2, 7/2 from $\gamma(\theta)$ in 1972Va07 and 1972Fr11. 2717.9γ M1+E2 to 3/2$^+$. $T_{1/2}$: from τ=100 fs *35* in 1972Fr11 by DSAM.
2935 *3*	(3/2)$^+$		J$^\pi$: spin=3/2, 5/2 from $\gamma(\theta)$ in 1972Va07 and 1975VaYG.
3421.0 *10*	5/2$^+$	<69 fs	J$^\pi$: spin=3/2, 5/2, 7/2 from $\gamma(\theta)$ in 1972Fr11. $T_{1/2}$: from τ<100 fs in 1972Fr11 by DSAM.
3558	(3/2$^-$,5/2$^+$)		E(level): rounded to the nearest integer based on the Adopted Levels. 1972Fr11 quotes 3563 *8* from 1969Mo12 without reporting their own E(level).
3592 *5*	7/2$^{(+)}$		
3802.0 *10*	3/2$^-$	25 fs *18*	J$^\pi$: spin=3/2 from $\gamma(\theta)$ in 1972Fr11. $T_{1/2}$: from τ=36 fs *26* in 1972Fr11 by DSAM.

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^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18 (continued)

^{35}S Levels (continued)

E(level)[†]	J$^\pi$[‡]	$T_{1/2}$	Comments
3815 *3*	(9/2)$^-$		
3885.0 *10*	(5/2$^+$)		
4025.0 *10*	3/2$^+$,5/2$^+$		
4107 *2*	(3/2$^-$,5/2$^+$)	<55 fs	$T_{1/2}$: from τ<80 fs in 1972Fr11 by DSAM.
4189 *2*	1/2$^-$	<35 fs	$T_{1/2}$: from τ<50 fs in 1972Fr11 by DSAM.
4480 *2*	(5/2$^+$)	<62 fs	$T_{1/2}$: from τ<90 fs in 1972Fr11 by DSAM.

† From the average of available measurements for the lowest three excited states, and from 1972Fr11 for higher-lying states.
‡ From the Adopted Levels. Supporting arguments from this dataset are given under comments if available.

γ(^{35}S)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.	δ	Comments
1572.33	1/2$^+$	1572.24 *17*	100	0	3/2$^+$			E_γ: from 1972Go31. A_2=−0.09 *5*, A_4=+0.07 *6* (1972Fr11); A_2=−0.007 *10*, A_4=+0.004 *20* (1975VaYG).
1992.9	7/2$^-$	1992.8	100	0	3/2$^+$	M2+E3	+0.17 *6*	Mult.: D+Q or Q+O from $\gamma(\theta)$ in 1970Bu18, 1972Fr11, 1972Va07, and 1975VaYG; E2+M3 is ruled out by RUL. D+Q is inconsistent with the J^π=7/2$^-$. δ: unweighted average of δ(E3/M2)=0.11 *4* (1970Bu18), +0.11 *4* (1972Fr11), and +0.28 *3* (1972Va07,1975VaYG) for J_i=7/2. Others: δ(Q/D)=−1.9 *7* (1970Bu18) and −0.43 *3* (1972Va07,1975VaYG) for J_i=5/2. A_2=+0.35 *4*, A_4=−0.39 *4* (1972Fr11); A_2=+0.16 *3*, A_4=−0.10 *4* (1975VaYG).
2348.8	3/2$^-$	776.5	27 *2*	1572.33	1/2$^+$	(E1)		Mult.,δ: D(+Q) from $\gamma(\theta)$, $\Delta\pi$=yes from level scheme. M2 component is less likely. δ<0.04 from $\gamma(\theta)$ and RUL in 1975VaYG, superseding 0<δ<+1.7 from $\gamma(\theta)$ in 1972Va07. A_2=−0.37 *3*, A_4=−0.03 *3* (1972Fr11); A_2=−0.38 *2*, A_4=+0.01 *4* (1975VaYG).
		2348.7	73 *2*	0	3/2$^+$	(E1)		Mult.,δ: D(+Q) from $\gamma(\theta)$, $\Delta\pi$=yes from level scheme. M2 component is less likely. δ<0.04 from $\gamma(\theta)$ and RUL in 1975VaYG, superseding −0.27 *12* from $\gamma(\theta)$ in 1972Va07; +0.01 *4* from $\gamma(\theta)$ in 1972Fr11 assuming pure E1 for 776.5γ. A_2=+0.33 *2*, A_4=−0.05 *2* (1972Fr11); A_2=+0.30 *2*, A_4=−0.02 *4* (1975VaYG).
2718.0	5/2$^+$	369.2	<5	2348.8	3/2$^-$			
		725.1	<5	1992.9	7/2$^-$			
		1145.7	<5	1572.33	1/2$^+$			
		2717.9	100	0	3/2$^+$	M1+E2	−0.39 *5*	Mult.,δ: D+Q from $\gamma(\theta)$ in 1975VaYG with −6.9<δ<−0.11 or +0.42<δ<+6.8 for J_i=3/2, −0.39 *5* or −15.0 *34* for J_i=5/2, 0.18 *4* or 3.8 *7* for J_i=7/2. E1+M2 is ruled out by RUL. δ=−0.39 *5* is adopted here based on J^π=5/2$^+$ in the Adopted Levels. −15.0 *34* is unlikely based on RUL. A_2=+0.47 *6*, A_4=−0.08 *7* (1972Fr11); A_2=+0.22 *2*, A_4=−0.02 *3* (1975VaYG).

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$^{34}S(d,p\gamma)$ 1972Fr11,1975VaYG,1970Bu18 (continued)

$\gamma(^{35}S)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Comments
2935	(3/2)$^+$	586	<10	2348.8	3/2$^-$	
		942	<20	1992.9	7/2$^-$	
		1363	<15	1572.33	1/2$^+$	
		2935	100	0	3/2$^+$	A_2=−0.06 *3*, A_4=−0.01 *6* (1975VaYG).
3421.0	5/2$^+$	486	<4	2935	(3/2)$^+$	
		703.0	<6	2718.0	5/2$^+$	
		1072.2	<7	2348.8	3/2$^-$	
		1428.1	<30	1992.9	7/2$^-$	
		1848.6	<7	1572.33	1/2$^+$	
		3420.8	100	0	3/2$^+$	A_2=−0.54 *8*, A_4=−0.11 *7* (1972Fr11).
3592	7/2$^{(+)}$	657	<10	2935	(3/2)$^+$	
		874	<11	2718.0	5/2$^+$	
		1243	<15	2348.8	3/2$^-$	
		1599	<17	1992.9	7/2$^-$	
		2020[#@]	<17	1572.33	1/2$^+$	
		3592	100	0	3/2$^+$	
3802.0	3/2$^-$	867	<4	2935	(3/2)$^+$	
		1084.0	5 *2*	2718.0	5/2$^+$	
		1453.2	8 *3*	2348.8	3/2$^-$	
		1809.1	<4	1992.9	7/2$^-$	
		2229.6	42 *4*	1572.33	1/2$^+$	A_2=−0.56 *15*, A_4=0.0 *1* (1972Fr11).
		3801.8	45 *4*	0	3/2$^+$	A_2=+0.19 *12*, A_4=0.0 *1* (1972Fr11).
3815	(9/2)$^-$	1822	100	1992.9	7/2$^-$	
3885.0	(5/2$^+$)	464.0	<30	3421.0	5/2$^+$	
		950	<12	2935	(3/2)$^+$	
		1167.0	<15	2718.0	5/2$^+$	
		1536.2	<18	2348.8	3/2$^-$	
		1892.1	100	1992.9	7/2$^-$	
		2312.6	<16	1572.33	1/2$^+$	
		3884.8	<16	0	3/2$^+$	
4025.0	3/2$^+$,5/2$^+$	467	<5	3558	(3/2$^-$,5/2$^+$)	
		604.0	<10	3421.0	5/2$^+$	
		1090	31 *8*	2935	(3/2)$^+$	
		1307.0	<8	2718.0	5/2$^+$	
		1676.2	33 *12*	2348.8	3/2$^-$	
		2032.0	<10	1992.9	7/2$^-$	
		2452.6	36 *12*	1572.33	1/2$^+$	
		4024.8	<13	0	3/2$^+$	
4107	(3/2$^-$,5/2$^+$)	549	<4	3558	(3/2$^-$,5/2$^+$)	
		686	<4	3421.0	5/2$^+$	
		1172	13 *6*	2935	(3/2)$^+$	
		1389	<10	2718.0	5/2$^+$	
		1758	<5	2348.8	3/2$^-$	
		2114	<7	1992.9	7/2$^-$	
		2535	<5	1572.33	1/2$^+$	
		4107	87 *6*	0	3/2$^+$	
4189	1/2$^-$	387[#@]	<4	3802.0	3/2$^-$	
		597[#@]	<4	3592	7/2$^{(+)}$	
		631	<4	3558	(3/2$^-$,5/2$^+$)	
		768[#@]	<4	3421.0	5/2$^+$	
		1254	<5	2935	(3/2)$^+$	
		1471[#@]	<5	2718.0	5/2$^+$	
		1840	48 *5*	2348.8	3/2$^-$	
		2196[#@]	<8	1992.9	7/2$^-$	

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^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18 (continued)

γ(^{35}S) (continued)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π
4189	1/2$^-$	2617	<6	1572.33	1/2$^+$
		4189	52 *5*	0	3/2$^+$
4480	(5/2$^+$)	595	<5	3885.0	(5/2$^+$)
		678	<5	3802.0	3/2$^-$
		888	<5	3592	7/2$^{(+)}$
		922	<5	3558	(3/2$^-$,5/2$^+$)
		1059	<10	3421.0	5/2$^+$
4480		1545	<7	2935	(3/2)$^+$
		1762	61 *7*	2718.0	5/2$^+$
		2131	<8	2348.8	3/2$^-$
		2487	<10	1992.9	7/2$^-$
		2908	39 *7*	1572.33	1/2$^+$
		4480	<12	0	3/2$^+$

† Deduced by evaluators from level-energy difference, unless otherwise noted.

‡ From 1972Fr11, unless otherwise noted.

γ transitions are considered questionable and not included in the Adopted Levels. The transition strengths estimated by evaluators exceed RUL. ^{34}S(n,γ) E=thermal observed other γ transitions from the same level but did not observe these questionable transitions.

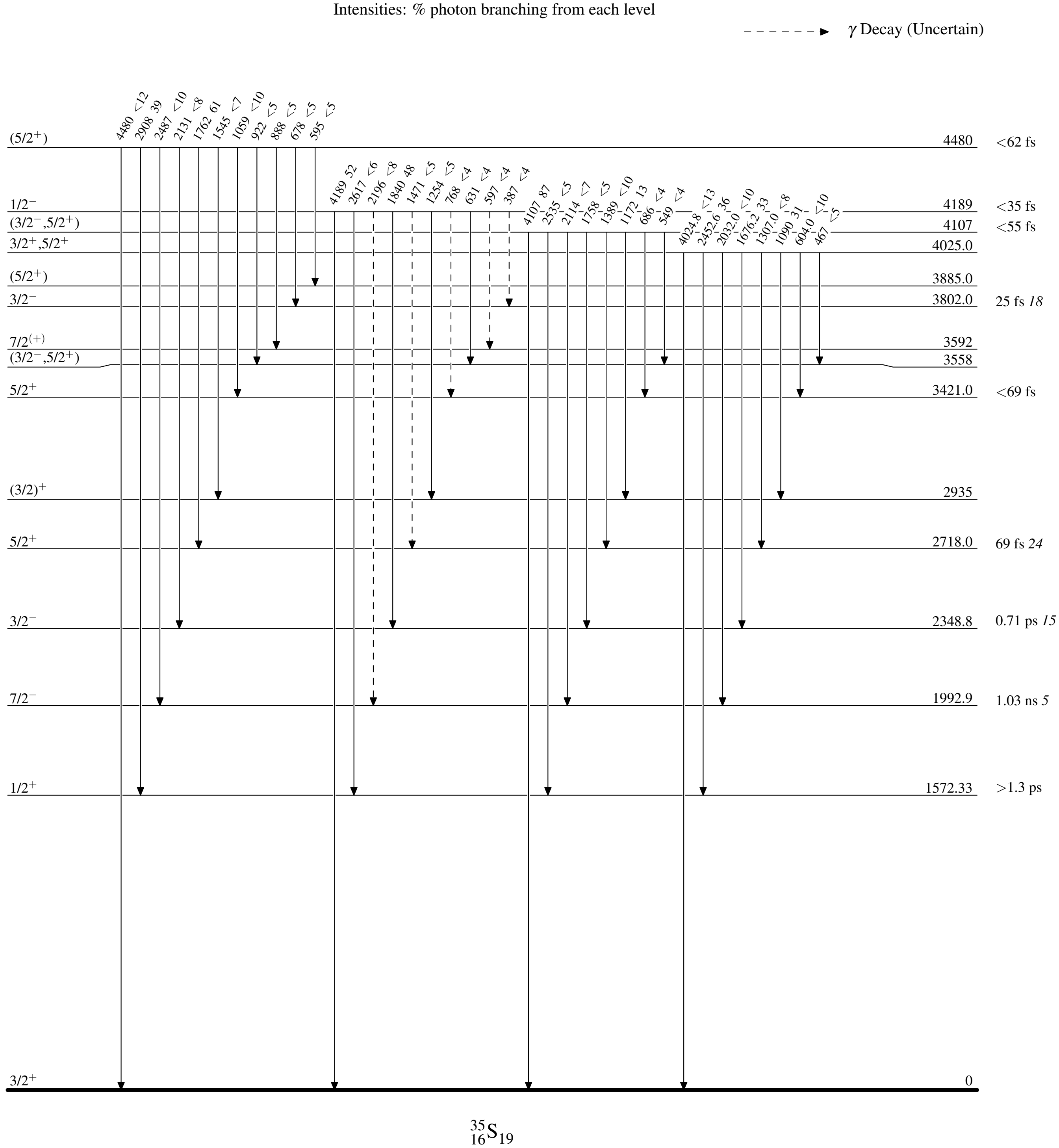
@ Placement of transition in the level scheme is uncertain.

^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18

Legend

Level Scheme

Intensities: % photon branching from each level

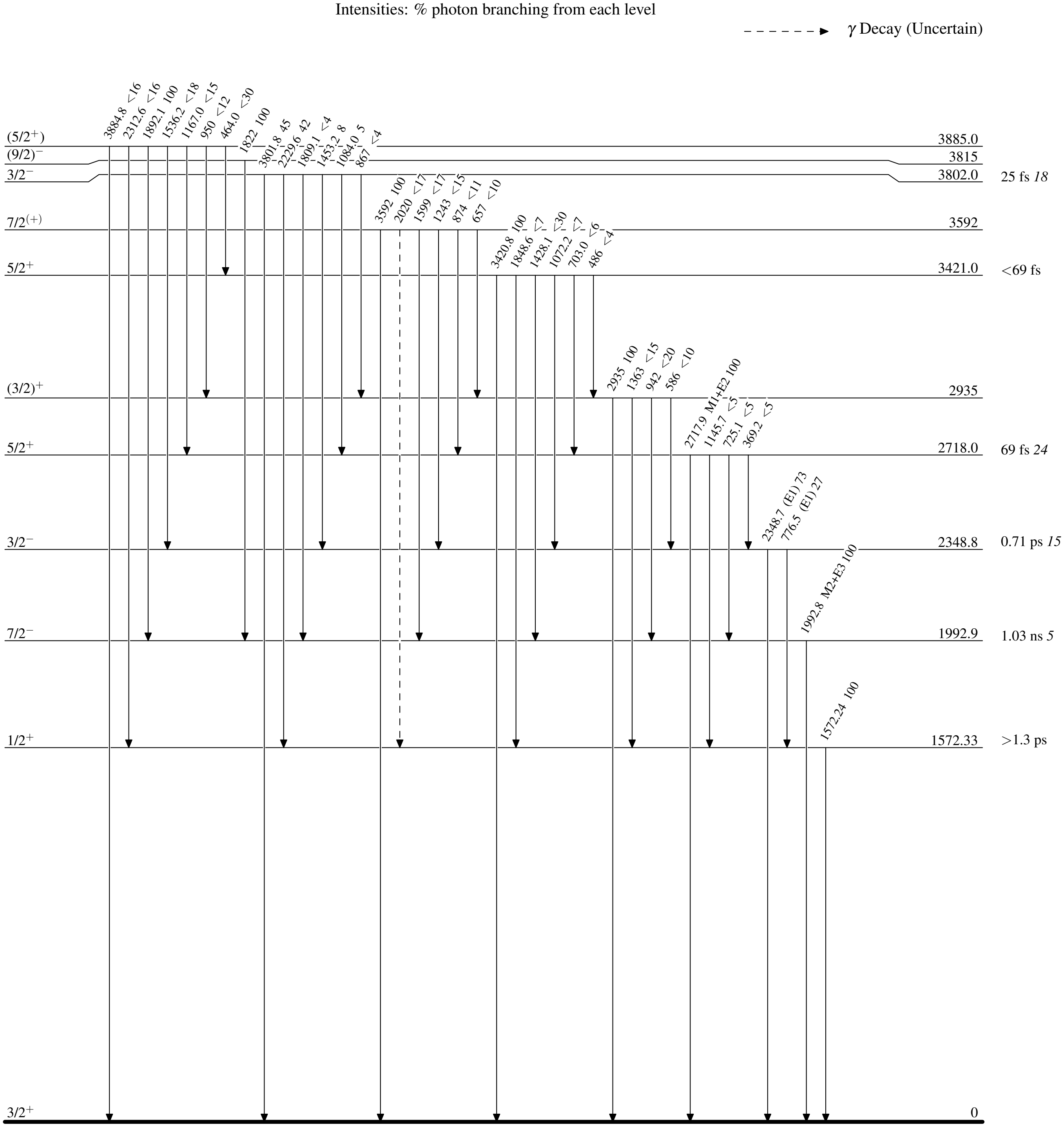


^{34}S(d,pγ) 1972Fr11,1975VaYG,1970Bu18

Legend

Level Scheme (continued)

Intensities: % photon branching from each level



$^{35}_{16}S_{19}$